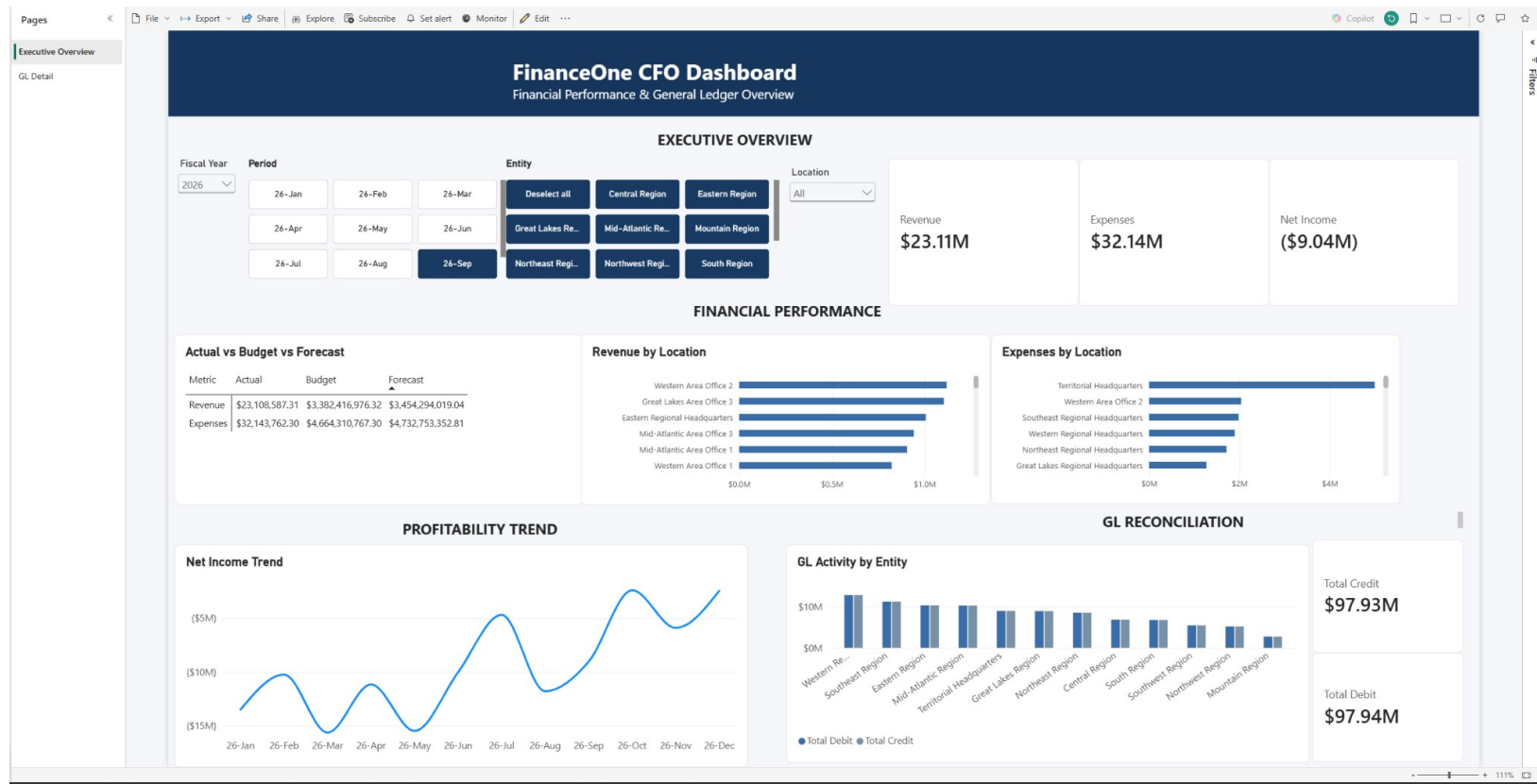


FinanceOne

Microsoft Fabric Financial Analytics Platform

End-to-End Financial Data Engineering, Semantic Modeling & Executive Reporting



Self-directed portfolio project using simulated ERP-style financial data.

Microsoft Fabric | OneLake | Lakehouse | Data Factory Pipelines | Dataflow Gen2 | Direct Lake | DAX | Power BI

1. Project Overview

Business challenge

Consolidate disconnected GL, budget, forecast, and reference data into one governed analytics platform.

Objective

Build an end-to-end Microsoft Fabric solution covering ingestion, medallion transformation, financial validation, dimensional modeling, Direct Lake, DAX, and executive reporting.

Scope

GL, chart of accounts, departments, entities, locations, fiscal calendar, budget, and forecast across FY2022-FY2026.

Project scale

60,000 GL journals

166,888 GL journal lines

269,406 Budget rows

5 years FY2022-FY2026

3 facts GL | Budget | Forecast

5 dimensions Account | Department | Entity | Fiscal Period
| Location

Solution Architecture

ERP Sources	Bronze	Silver	Gold	Direct Lake	Power BI
GL Budget Forecast Reference	Raw Delta	Transform + Validate	Dimensions + Facts	Semantic Model	CFO + GL Detail

Design principle: Raw source fidelity in Bronze; standardized and validated data in Silver; conformed dimensions/facts in Gold; Direct Lake semantic model for Power BI consumption.

2. Fabric Workspace & Orchestration

FinanceOne-ERP-DEV Development workspace for the FinanceOne ERP Analytics Platform. This solution ingests Sage Intacct and Microsoft Dynamics-style financial data ...				
+ New item New folder Import Migrate				
	Name	Status	Type	Task
	DF_Budget_Bronze_To_Silver		Dataflow Ge...	—
	DF_Budget_Silver_To_Gold		Dataflow Ge...	—
	DF_Forecast_Bronze_To_Silver		Dataflow Ge...	—
	DF_Forecast_Silver_To_Gold		Dataflow Ge...	—
	DF_GL_Bronze_To_Silver		Dataflow Ge...	—
	DF_Reference_Bronze_To_Silver		Dataflow Ge...	—
	DF_Silver_To_Gold_Finance		Dataflow Ge...	—
	FinanceOne_Bronze_LH		Lakehouse	—
	FinanceOne_Bronze_LH		SQL analytics...	—
	FinanceOne_CFO_Dashboard		Report	—
	FinanceOne_Finance_Model		Semantic mo...	—
	FinanceOne_Gold_LH		Lakehouse	—
	FinanceOne_Gold_LH		SQL analytics...	—
	FinanceOne_Model_Validation		Report	—
	FinanceOne_Silver_LH		Lakehouse	—
	FinanceOne_Silver_LH		SQL analytics...	—
	PL_ERP_Budget_To_Bronze		Pipeline	—
	PL_ERP_Forecast_Bronze_Ingestion		Pipeline	—
	PL_ERP_GL_Bronze_Ingestion		Pipeline	—
	PL_ERP_Reference_Bronze_Ingestion		Pipeline	—
	PL_FinanceOne_End_To_End_Refresh		Pipeline	—

FinanceOne-ERP-DEV workspace with pipelines, dataflows, Lakehouses, semantic model, and reports.

Final workspace design

- Bronze, Silver, and Gold Lakehouses.
- Reference, GL, Budget, and Forecast ingestion pipelines.
- Bronze-to-Silver and Silver-to-Gold Dataflow Gen2 transformations.
- PL_FinanceOne_End_To_End_Refresh for orchestration.
- FinanceOne_Finance_Model and final CFO/GL Detail reporting.
- Workspace organization demonstrates a complete engineering solution rather than a standalone Power BI report.

3. Data Ingestion & Bronze Layer

ERP-style source files are organized in OneLake and loaded into Bronze Delta tables. The ingestion layer is independently rerunnable and monitored before transformation.

	budget_id	fiscal_year	fiscal_period	entity_code	account_number	department_code	budget_amount
1	BUD000001	2022	1	CENTRAL	6000	FIN	155622.47
2	BUD000002	2022	2	CENTRAL	6000	FIN	154838.84
3	BUD000003	2022	3	CENTRAL	6000	FIN	151976.87
4	BUD000004	2022	4	CENTRAL	6000	FIN	152923
5	BUD000005	2022	5	CENTRAL	6000	FIN	151752.12
6	BUD000006	2022	6	CENTRAL	6000	FIN	150824.16
7	BUD000007	2022	7	CENTRAL	6000	FIN	150461.06
8	BUD000008	2022	8	CENTRAL	6000	FIN	150912.62
9	BUD000009	2022	9	CENTRAL	6000	FIN	154212.07
10	BUD000010	2022	10	CENTRAL	6000	FIN	152276.15
11	BUD000011	2022	11	CENTRAL	6000	FIN	164005.89
12	BUD000012	2022	12	CENTRAL	6000	FIN	163965.1
13	BUD000013	2022	1	CENTRAL	6110	FIN	4694.88
14	BUD000014	2022	2	CENTRAL	6110	FIN	4648.37
15	BUD000015	2022	3	CENTRAL	6110	FIN	4751.7
16	BUD000016	2022	4	CENTRAL	6110	FIN	4622.76
17	BUD000017	2022	5	CENTRAL	6110	FIN	4696.06
18	BUD000018	2022	6	CENTRAL	6110	FIN	4712.15
19	BUD000019	2022	7	CENTRAL	6110	FIN	4632.47
20	BUD000020	2022	8	CENTRAL	6110	FIN	4626.73
21	BUD000021	2022	9	CENTRAL	6110	FIN	4633.76
22	BUD000022	2022	10	CENTRAL	6110	FIN	4632.21
23	BUD000023	2022	11	CENTRAL	6110	FIN	4740.08
24	BUD000024	2022	12	CENTRAL	6110	FIN	5205.88
25	BUD000025	2022	1	CENTRAL	6140	FIN	2876.24
26	BUD000026	2022	2	CENTRAL	6140	FIN	2873.33
27	BUD000027	2022	3	CENTRAL	6140	FIN	2887.44
28	BUD000028	2022	4	CENTRAL	6140	FIN	2875.34
29	BUD000029	2022	5	CENTRAL	6140	FIN	2889.9
30	BUD000030	2022	6	CENTRAL	6140	FIN	3054.49
31	BUD000031	2022	7	CENTRAL	6140	FIN	3016.29
32	BUD000032	2022	8	CENTRAL	6140	FIN	2847.33
33	BUD000033	2022	9	CENTRAL	6140	FIN	2826.71
34	BUD000034	2022	10	CENTRAL	6140	FIN	3031.06
35	BUD000035	2022	11	CENTRAL	6140	FIN	2915.1
36	BUD000036	2022	12	CENTRAL	6140	FIN	3156.07
37	BUD000037	2022	1	CENTRAL	6150	FIN	15205.37
38	BUD000038	2022	2	CENTRAL	6150	FIN	15487.4
39	BUD000039	2022	3	CENTRAL	6150	FIN	12773.53

Bronze Lakehouse with source folders and raw financial tables.

Key implementation points: separate domain pipelines, source-to-Bronze copy activities, monitored execution, and repeatable downstream handoff.

Copy data

Copy_GL_JournalLines_To_Bronze

Parameters Variables Settings Output Library variables

Pipeline run ID: d174b718-605b-4d1f-924f-6edc2aeb7bb Pipeline status: Succeeded

Activity name: Copy_GL_JournalLines_To_Bronze Activity status: Succeeded Run start: 9/7/2025, 5:41:27 PM Duration: 23s Input: Output:

Successful GL journal-line ingestion run - 23 seconds.

4. Silver Transformation & Financial Data Quality

Dataflow Gen2 and Power Query standardize journal data, enforce datatypes, align dimensional keys, and implement finance-specific validation before promotion to Gold.

Power Query interface showing a table of journal data. The table has columns for journal ID, date, fiscal year, period, entity, department, location, journal number, account number, journal type, description, debit amount, credit amount, source system, posted status, and reference number. The data is filtered for 'silver_gl_journal_lines' and 'silver_gl_journal_balance'.

Journal ID	Journal Date	Fiscal Year	Fiscal Period	Entity	Department	Location	Journal Number	Account Number	Journal Type	Journal Description	1.2 debit amount	1.2 credit amount	Source System	Posted Status	Reference Number
1 JRN-2024-000001	3/22/2024	2024	3	MIDATL	SALES	MA-004	1	1100	SALE	Product sale	17729.21	0	ERP	POSTED	ERP-701505
2 JRN-2024-000001	3/22/2024	2024	3	MIDATL	SALES	MA-004	2	4000	SALE	Product sale	17729.21	0	ERP	POSTED	ERP-701505
3 JRN-2024-000001	3/22/2024	2024	3	MIDATL	SALES	MA-001	3	5010	SALE	Product sale	5663.71	0	ERP	POSTED	ERP-701505
4 JRN-2024-000001	3/22/2024	2024	3	MIDATL	SALES	MA-004	4	1200	SALE	Product sale	0	5663.71	ERP	POSTED	ERP-701505
5 JRN-2026-000002	3/25/2026	2026	3	MOUNTAIN	HR	MTN-001	1	1310	PREPAID	Prepaid purchase	23090.54	0	ERP	POSTED	ERP-201304
6 JRN-2026-000002	3/25/2026	2026	3	MOUNTAIN	FIN	MTN-001	2	2000	PREPAID	Prepaid purchase	0	23090.54	ERP	POSTED	ERP-201304
7 JRN-2023-000003	8/1/2023	2023	8	WEST	OPS	WEST-004	1	1000	CASH_RECEIPT	Customer cash receipt	122045.17	0	Manual	POSTED	MAN-346557
8 JRN-2023-000003	8/1/2023	2023	8	WEST	OPS	WEST-002	2	1100	CASH_RECEIPT	Customer cash receipt	0	122045.17	Manual	POSTED	MAN-346557
9 JRN-2025-000004	12/27/2025	2025	12	CENTRAL	FIN	CENT-001	1	1200	OTHER	Non-operating income	36796.11	0	Billing	POSTED	BL-459177
10 JRN-2026-000004	12/27/2025	2025	12	CENTRAL	FIN	CENT-001	2	8030	OTHER	Non-operating income	0	36796.11	Billing	POSTED	BL-459177
11 JRN-2024-000005	9/13/2024	2024	9	MIDATL	FIN	MA-004	1	6000	PAYROLL	Payroll posting	63910.29	0	ERP	POSTED	ERP-831167
12 JRN-2024-000005	9/13/2024	2024	9	MIDATL	SALES	MA-004	2	6110	PAYROLL	Payroll posting	10082.18	0	ERP	POSTED	ERP-831167
13 JRN-2024-000005	9/13/2024	2024	9	MIDATL	MKTG	MA-004	3	2200	PAYROLL	Payroll posting	0	1230.25	ERP	POSTED	ERP-831167
14 JRN-2024-000005	9/13/2024	2024	9	MIDATL	HR	MA-003	4	2210	PAYROLL	Payroll posting	0	738.15	ERP	POSTED	ERP-831167
15 JRN-2024-000005	9/13/2024	2024	9	MIDATL	MAINT	MA-001	5	2220	PAYROLL	Payroll posting	0	2214.44	ERP	POSTED	ERP-831167
16 JRN-2024-000005	9/13/2024	2024	9	MIDATL	HR	MA-001	6	2230	PAYROLL	Payroll posting	0	738.15	ERP	POSTED	ERP-831167
17 JRN-2024-000005	9/13/2024	2024	9	MIDATL	MAINT	MA-003	7	2250	PAYROLL	Payroll posting	0	10282.18	ERP	POSTED	ERP-831167
18 JRN-2024-000005	9/13/2024	2024	9	MIDATL	OPS	MA-001	8	1000	PAYROLL	Payroll posting	0	60863.3	ERP	POSTED	ERP-831167
19 JRN-2026-000006	12/05/2026	2026	12	NORTHEAST	HR	NE-004	1	2000	AP_PAYMENT	Vendor payment	134586.21	0	ERP	POSTED	ERP-793296
20 JRN-2026-000006	12/05/2026	2026	12	NORTHEAST	FIN	NE-001	2	1000	AP_PAYMENT	Vendor payment	0	134586.21	ERP	POSTED	ERP-793296
21 JRN-2023-000007	4/23/2023	2023	4	MOUNTAIN	HR	MTN-001	1	1550	ASSET_PURCHASE	Capital asset purchase	114176.1	0	ERP	POSTED	ERP-329666
22 JRN-2023-000007	4/23/2023	2023	4	MOUNTAIN	HR	MTN-001	2	2000	ASSET_PURCHASE	Capital asset purchase	0	114176.1	ERP	POSTED	ERP-329666
23 JRN-2023-000008	2/1/2025	2025	2	MIDATL	MAINT	MA-002	1	6000	PAYROLL	Payroll posting	293629.56	0	AP	POSTED	AP-589972
24 JRN-2023-000008	2/1/2025	2025	2	MIDATL	OPS	MA-003	2	6150	PAYROLL	Payroll posting	42594.26	0	AP	POSTED	AP-589972
25 JRN-2023-000008	2/1/2025	2025	2	MIDATL	HR	MA-004	3	2200	PAYROLL	Payroll posting	0	5915.67	AP	POSTED	AP-589972
26 JRN-2023-000008	2/1/2025	2025	2	MIDATL	IT	MA-001	4	2210	PAYROLL	Payroll posting	0	3549.4	AP	POSTED	AP-589972
27 JRN-2023-000008	2/1/2025	2025	2	MIDATL	MAINT	MA-001	5	2220	PAYROLL	Payroll posting	0	10648.2	AP	POSTED	AP-589972
28 JRN-2023-000008	2/1/2025	2025	2	MIDATL	HR	MA-001	6	2230	PAYROLL	Payroll posting	0	3549.4	AP	POSTED	AP-589972
29 JRN-2023-000008	2/1/2025	2025	2	MIDATL	OPS	MA-002	7	2250	PAYROLL	Payroll posting	0	42594.26	AP	POSTED	AP-589972
30 JRN-2023-000008	2/1/2025	2025	2	MIDATL	MKTG	MA-001	8	1000	PAYROLL	Payroll posting	0	26996.88	AP	POSTED	AP-589972
31 JRN-2026-000009	3/2/2026	2026	3	SOUTHWEST	HR	SW-001	1	2000	AP_PAYMENT	Vendor payment	108005.27	0	Payroll	POSTED	PAW-923667
32 JRN-2026-000009	3/2/2026	2026	3	SOUTHWEST	HR	SW-001	2	1000	AP_PAYMENT	Vendor payment	0	108005.27	Payroll	POSTED	PAW-923667
33 JRN-2023-000010	3/5/2023	2023	3	GREATLAKES	SALES	GL-003	1	1000	CASH_RECEIPT	Customer cash receipt	139031.58	0	AP	POSTED	AP-557812
34 JRN-2023-000010	3/5/2023	2023	3	GREATLAKES	SALES	GL-002	2	1100	CASH_RECEIPT	Customer cash receipt	0	139031.58	AP	POSTED	AP-557812
35 JRN-2023-000011	8/3/2025	2025	8	MIDATL	SALES	MA-004	1	1100	SALE	Recurring service billing	87155.79	0	AP	POSTED	AP-936080
36 JRN-2023-000011	8/3/2025	2025	8	MIDATL	SALES	MA-003	2	4310	SALE	Recurring service billing	0	87155.79	AP	POSTED	AP-936080
37 JRN-2026-000012	11/21/2026	2026	11	CENTRAL	FIN	CENT-002	1	2000	AP_PAYMENT	Vendor payment	4686.12	0	Manual	POSTED	MAN-721307
38 JRN-2026-000012	11/21/2026	2026	11	CENTRAL	FIN	CENT-001	2	1000	AP_PAYMENT	Vendor payment	0	4686.12	Manual	POSTED	MAN-721307
39 JRN-2026-000013	8/10/2026	2026	8	SOUTHWEST	HR	SW-001	1	7140	DEPRECIATION	Monthly depreciation	26326.1	0	Billing	POSTED	BL-911662
40 JRN-2026-000013	8/10/2026	2026	8	SOUTHWEST	HR	SW-001	2	1640	DEPRECIATION	Monthly depreciation	0	26326.1	Billing	POSTED	BL-911662
41 JRN-2023-000014	10/31/2023	2023	10	MIDATL	MKTG	MA-001	1	6900	VENDOR_INVOICE	Admin vendor invoice	4206.87	0	ERP	POSTED	ERP-702303
42 JRN-2023-000014	10/31/2023	2023	10	MIDATL	HR	MA-001	2	2000	VENDOR_INVOICE	Admin vendor invoice	0	4206.87	ERP	POSTED	ERP-702303
43 JRN-2026-000015	8/14/2026	2026	8	EAST	OPS	EAST-001	1	1100	SALE	Professional services billing	104317.05	0	AP	POSTED	AP-374208
44 JRN-2026-000015	8/14/2026	2026	8	EAST	OPS	EAST-001	2	4110	SALE	Professional services billing	0	104317.05	AP	POSTED	AP-374208
45 JRN-2024-000016	3/5/2024	2024	3	MIDATL	SALES	MA-003	1	1000	CASH_RECEIPT	Customer cash receipt	29172.82	0	ERP	POSTED	ERP-959600
46 JRN-2024-000016	3/5/2024	2024	3	MIDATL	SALES	MA-003	2	1100	CASH_RECEIPT	Customer cash receipt	0	29172.82	ERP	POSTED	ERP-959600
47 JRN-2023-000017	5/21/2023	2023	5	SOUTHEAST	MAINT	SE-001	1	6300	VENDOR_INVOICE	Repairs vendor invoice	3524.24	0	Payroll	POSTED	PAW-736482
48 JRN-2023-000017	5/21/2023	2023	5	SOUTHEAST	MAINT	SE-004	2	2000	VENDOR_INVOICE	Repairs vendor invoice	0	3524.24	Payroll	POSTED	PAW-736482
49 JRN-2023-000018	7/27/2025	2025	7	EAST	MKTG	EAST-004	1	6000	PAYROLL	Payroll posting	304346.68	0	ERP	POSTED	ERP-736247

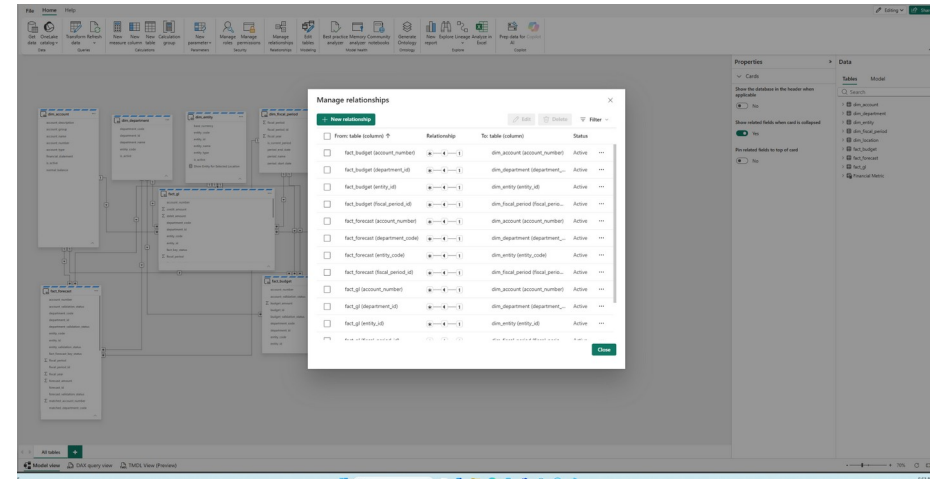
DF_GL_Bronze_To_Silver - journal fields, source-system attributes, amounts, applied steps, and separate journal-balance validation query.

- Standardized account, department, entity, location, and fiscal-period attributes.
- Validated missing/invalid dimensional keys and controlled Bronze exceptions.
- Maintained debit/credit detail and created signed financial logic.
- Used journal-balance checks to verify debit/credit integrity.

5. Gold Dimensional Model & Direct Lake Semantic Layer

[illegible]

Gold Lakehouse - conformed dimensions and fact tables.



Semantic model - active many-to-one finance relationships.

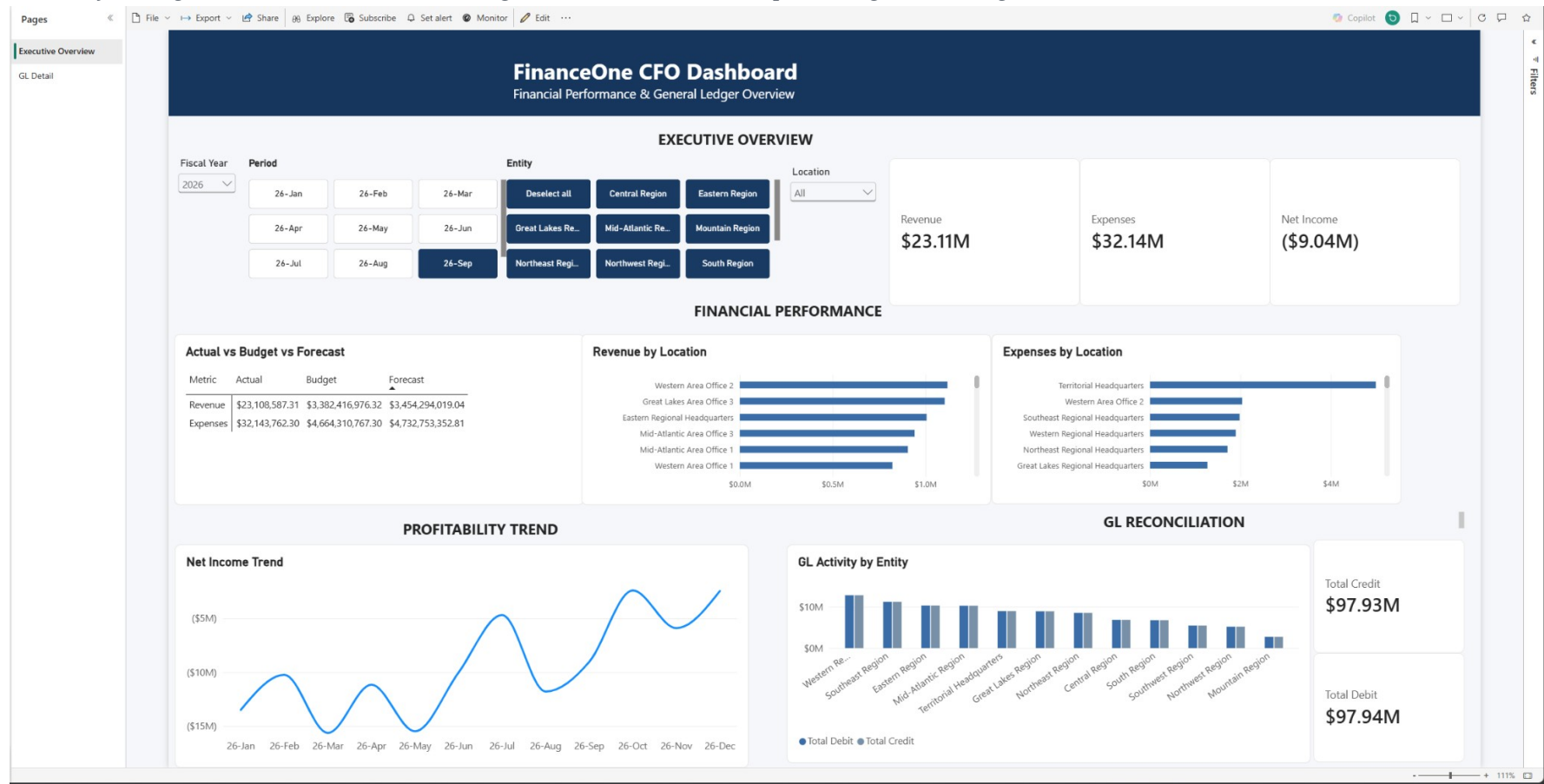
- Dimensions: Account, Department, Entity, Fiscal Period, Location.
- Facts: General Ledger, Budget, Forecast.
- Direct Lake semantic model for analytics consumption.

- Resolved relationship-key datatype mismatches.
- Avoided ambiguous physical relationship paths.
- Used disconnected Financial Metric table for measure-driven comparisons.

Selected DAX: Net Amount, Revenue, Expenses, Actual, Net Income, Budget/Forecast comparisons, plus helper measures for reciprocal Entity/Location slicer behavior.

6. CFO Executive Dashboard

The executive layer integrates actual GL results with budget and forecast data while preserving drillable organizational context.



Executive Overview | Financial Performance | Profitability Trend | GL Reconciliation

- Revenue, Expenses, and Net Income KPIs.
- Actual vs Budget vs Forecast comparison.
- Revenue and Expenses by Location.
- Full-year Net Income trend remains visible under single-period selection.
- GL Activity by Entity plus Total Debit and Total Credit reconciliation.

7. General Ledger Detail & Reconciliation

The GL Detail page provides the audit trail behind executive reporting and supports transaction-level investigation by fiscal year, period, entity, location, and account.

Pages<<FileExportShareExploreSubscribeSet alertMonitorEdit<<<Copilot

Executive OverviewGL Detail

Filters

FinanceOne — GL Detail
General Ledger Transaction Detail & Reconciliation

TRANSACTION FILTERS

Fiscal Year2026

Period

Entity

Location

Account

26-Jan26-Feb

26-Mar26-Apr26-May

26-Jun26-Jul26-Aug

Deselect...Central R...Eastern R...Great Lak...Mid-Atlan...Mountain...Northeast...Northwes...South Re...

Deselect...Accountin... 6510Accounts ... 2000Accounts ... 1100Accrued ... 2110Accrued P... 2100Accrued P... 2120Accrued ... 2130Accumula... 1720

GENERAL LEDGER DETAIL

Journal ID	Entity	Account	Department	Location	Description	Debit	Credit
JRN-2026-000045	SOUTHWEST	1000	OPS	SW-002	Customer cash receipt	10,966.26	0.00
JRN-2026-000045	SOUTHWEST	1100	SALES	SW-001	Customer cash receipt	0.00	10,966.26
JRN-2026-000083	CENTRAL	1000	SALES	CENT-001	Customer cash receipt	75,101.19	0.00
JRN-2026-000083	CENTRAL	1100	BADDEPT	CENT-003	Customer cash receipt	0.00	75,101.19
JRN-2026-000226	WEST	1300	FIN	WEST-004	Prepaid purchase	94,295.35	0.00
JRN-2026-000226	WEST	2000	HR	WEST-001	Prepaid purchase	0.00	94,295.35
JRN-2026-000251	NORTHWEST	1330	HR	NW-003	Prepaid purchase	32,296.93	0.00
JRN-2026-000251	NORTHWEST	2000	HR	NW-002	Prepaid purchase	0.00	32,296.93
JRN-2026-000267	SOUTHEAST	1000	HR	SE-001	Vendor payment	0.00	140,071.92
JRN-2026-000267	SOUTHEAST	2000	FIN	SE-001	Vendor payment	140,071.92	0.00
JRN-2026-000533	NORTHWEST	1000	OPS	NW-002	Customer cash receipt	36,448.38	0.00
JRN-2026-000533	NORTHWEST	1100	SALES	NW-003	Customer cash receipt	0.00	36,448.38
Total						97,936,077.13	97,932,483.68

GL RECONCILIATION

Total Debit
\$97.94M

Total Credit
\$97.93M

Net Amount
\$3.59K

Journal ID | Entity | Account | Department | Location | Description | Debit | Credit | Reconciliation cards

6:01 PM 9/7/2026

- Transaction-level audit trail and dimensional filtering.
- Reciprocal Entity/Location slicer logic without ambiguous model relationships.
- Total Debit, Total Credit, and Net Amount provide immediate reconciliation context.
- Consistent styling and filter behavior with the CFO dashboard.

8. Engineering Challenges Solved

Stale destination schemas

Recreated affected Silver/Gold destinations and reran dataflows.

Direct Lake datatype mismatches

Standardized relationship keys to compatible text datatypes.

Ambiguous model paths

Preserved the star model and avoided a conflicting Entity-to-Location relationship.

Dependent slicers

Used DAX helper measures for Entity-to-Location and Location-to-Entity behavior.

Fiscal period behavior

Sorted period labels by numeric fiscal period and controlled visual interactions.

GL visual schema changes

Repaired the GL Detail visual after source/model field changes.

Financial validation

Validated debit/credit balance, dimensional keys, and controlled source exceptions.

End-to-end maintainability

Separated ingestion, transformation, Gold curation, modeling, and reporting into testable artifacts.

Technology Stack

Microsoft Fabric | OneLake | Lakehouse | Data Factory Pipelines | Dataflow Gen2 | Power Query | Direct Lake | Semantic Models | DAX | Power BI

Project Outcome

FinanceOne demonstrates an end-to-end financial analytics implementation that combines finance-domain controls with cloud data engineering: ingestion, medallion transformation, dimensional modeling, semantic design, DAX, executive reporting, and transaction-level reconciliation.